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Prelab 3: Electric Field Mapping

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6. a. What do you observe about the geometric relationship between the electric field vector and the equipotential lines?

   The electric field vector always points perpendicular to the equipotential lines.

   b. Calculate the magnitude of the electric field vectors at at least three points on your grid.

   - We know that the magnitude of electric field due to a point charge q is $E = \frac{kq}{r^2}$.
   - Since $V = -\int \vec{E} \cdot \mathrm{d}\vec{r}$, we can deduce that $|V| = \left|-\int E\,\mathrm{d}r\right| = \left|-\int \frac{kq}{r^2}\,\mathrm{d}r\right| = \frac{kq}{r}$.
   - Thus, we can rewrite the formula for the magnitude of electric field as $E = \frac{V}{r}$.

   Here were the three points I chose:

   1. At $r = 100\,\text{cm} = 1.00\,\text{m}$, $V = 9.017\,\text{V}$ so $E = \frac{V}{r} = \frac{9.017\,\text{V}}{1.00\,\text{m}} = 9.017\,\text{V}\,\text{m}^{-1}$.
   2. At $r = 158.5\,\text{cm} = 1.585\,\text{m}$, $V = 5.674\,\text{V}$ so $E = \frac{V}{r} = \frac{5.674\,\text{V}}{1.585\,\text{m}} = 3.579\,\text{V}\,\text{m}^{-1}$.
   3. At $r = 211.8\,\text{cm} = 2.118\,\text{m}$, $V = 4.258\,\text{V}$ so $E = \frac{V}{r} = \frac{4.258\,\text{V}}{2.118\,\text{m}} = 2.010\,\text{V}\,\text{m}^{-1}$.

7. How does the electric field of 1 positive and 1 negative charge compare with the field from a single positive charge?

   Since $\vec{E} = -\vec{\nabla} V = -\left\langle \frac{\partial V}{\partial x}, \frac{\partial V}{\partial y} \right\rangle$, the magnitude $|E| = \sqrt{\left(\frac{\partial V}{\partial x}\right)^2 + \left(\frac{\partial V}{\partial y}\right)^2}$.

   We can use the approximations $\frac{\partial V}{\partial x} \approx \frac{V_2 - V_1}{\Delta x}$ and $\frac{\partial V}{\partial y} \approx \frac{V_2 - V_1}{\Delta y}$ to get

   $$|E| \approx \sqrt{\left(\frac{V_2 - V_1}{\Delta x}\right)^2 + \left(\frac{V_2 - V_1}{\Delta y}\right)^2}.$$

   1. The electric field perfectly in the center of the two charges is $E = \frac{2kq}{r^2}$ where r is the distance from the center to each charge and $q = 1\,\text{nC}$. Using the ruler tool, $r = 75\,\text{cm} = 0.75\,\text{m}$, which makes $E \approx 32\,\text{N}\,\text{C}^{-1}$.
   2. For two points with distance components $(\Delta x, \Delta y) = (0.2\,\text{m}, 0.2\,\text{m})$, we have $V_1 = -3.981\,\text{V}$ and $V_2 = -2.867\,\text{V}$. Plugging these values in, we get $E \approx 7.88\,\text{N}\,\text{C}^{-1}$.
   3. At another location with the same Δx and Δy, I measured $V_1 = 13.63\,\text{V}$ and $V_2 = 19.79\,\text{V}$, which gives us $E \approx 41.1\,\text{N}\,\text{C}^{-1}$.

   The electric field shoots out from the positive charge and curves towards the negative charge in all directions.

8. How does the electric field of a vertical line of positive charges and a negative charge at some distance from the line charge compare with the field from a single positive charge?

   The electric field shooting out from the line of positive charges is dominant, and as I increase the length of the line charge, the electric field becomes more uniform, stronger, and straight. There is a slight curve in towards the negative charge, especially when we are near it.